

Table 1: Sequential MAS results for Qwen3-8B. Acc. is accuracy (%), Single, TextMAS, and Linear times are converted to the Kernel machine’s throughput; original A100 Speed values are shown in parentheses. The remaining methods retain their H200 measurements. Best results in each row are bold and second-best results are underlined. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by —.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
Sequential MAS Setting									
AIME24	Acc.	50.0	53.3	<u>64.58</u>	56.7	66.25	64.17	63.75	↑ 13.0
	Token	12891	38596	<u>8607</u>	8953	8548	8770	22342	↓ 77.9%
	Time	340.2 (421)	1018.6 (2808)	229.8	236.3 (688)	225.6	<u>225.8</u>	549.0	× 4.5
AIME25	Acc.	46.7	53.3	48.75	53.3	49.58	<u>50.42</u>	49.17	↓ 2.9
	Token	14692	45088	9885	8699	10158	9830	24557	↓ 78.2%
	Time	377.8 (450)	1159.4 (3150)	256.0	223.7 (820)	261.2	<u>255.5</u>	597.0	× 4.5
ARC-C	Acc.	91.0	94.6	94.54	94.4	<u>94.80</u>	95.56	94.45	↑ 1.0
	Token	846	2252	<u>559</u>	529	570	915	12970	↓ 74.7%
	Time	5.6 (266)	14.8 (2059)	<u>3.7</u>	3.5 (703)	<u>3.7</u>	4.5	65.3	× 4.0
ARC-E	Acc.	95.6	99.1	98.27	<u>98.8</u>	98.32	98.40	97.94	↓ 0.7
	Token	656	2085	468	490	472	822	10190	↓ 77.4%
	Time	3.5 (404)	11.0 (3702)	2.5	<u>2.6 (1759)</u>	2.5	3.2	48.8	× 4.4
GPQA	Acc.	39.9	43.4	49.87	45.5	48.74	48.61	<u>49.24</u>	↑ 5.3
	Token	6435	17986	3562	4571	<u>3588</u>	3846	13076	↓ 80.1%
	Time	104.6 (813)	292.2 (5771)	57.5	74.3 (854)	<u>58.3</u>	59.4	257.6	× 5.0
GSM8K	Acc.	81.1	92.3	91.58	93.8	91.74	92.19	—	↓ 0.1
	Token	1280	2324	628	860	642	1029	—	↓ 72.4%
	Time	7.9 (449)	14.3 (1739)	4.1	5.3 (543)	4.1	<u>5.2</u>	—	× 3.5
HumanEval+	Acc.	74.4	80.5	84.15	80.5	<u>83.54</u>	83.38	82.62	↑ 3.0
	Token	2507	4593	<u>1668</u>	1866	1637	1696	15055	↓ 64.4%
	Time	31.5 (502)	57.7 (1619)	<u>18.9</u>	23.4 (497)	18.8	21.5	112.4	× 3.1
MBPP+	Acc.	64.8	69.5	—	74.6	75.26	<u>75.20</u>	73.15	↑ 5.8
	Token	2053	3695	—	1164	<u>1462</u>	1528	14388	↓ 60.4%
	Time	28.9 (1064)	52.1 (3628)	—	16.4 (1275)	<u>17.9</u>	19.6	101.6	× 2.9
MedQA	Acc.	53.0	75.0	76.33	<u>75.3</u>	73.67	72.00	66.67	↓ 1.3
	Token	2098	4260	1872	1555	1139	<u>1154</u>	15972	↓ 73.3%
	Time	18.3 (476)	37.1 (1923)	17.0	<u>13.5 (928)</u>	12.5	13.6	111.1	× 3.0

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 2: Hierarchical MAS results for Qwen3-8B. Acc. is accuracy (%), Single, TextMAS, and Linear times are converted to the Kernel machine’s throughput; original A100 Speed values are shown in parentheses. The remaining methods retain their H200 measurements. Best results in each row are bold and second-best results are underlined. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by –.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
<i>Hierarchical MAS Setting</i>									
AIME24	Acc.	50.0	53.3	—	53.3	<u>60.00</u>	65.83	—	↑ 12.5
	Token	12891	42629	—	7526	9926	<u>9898</u>	—	↓ 76.8%
	Time	335.6 (421)	1109.7 (3132)	—	195.9 (776)	258.4	<u>257.9</u>	—	× 4.3
AIME25	Acc.	46.7	50.0	—	50.0	53.33	<u>52.50</u>	—	↑ 3.3
	Token	14692	53929	—	13230	11207	<u>11300</u>	—	↓ 79.2%
	Time	379.0 (450)	1391.2 (3488)	—	341.3 (616)	289.1	<u>291.9</u>	—	× 4.8
ARC-C	Acc.	91.0	93.3	<u>94.20</u>	93.9	95.05	95.05	—	↑ 1.8
	Token	846	2854	606	344	603	822	—	↓ 78.9%
	Time	5.3 (266)	17.9 (2034)	3.9	2.2 (714)	<u>3.8</u>	4.3	—	× 4.7
ARC-E	Acc.	95.6	98.2	98.48	<u>98.3</u>	98.23	98.23	—	↑ 0.0
	Token	656	2237	<u>494</u>	308	495	721	—	↓ 77.9%
	Time	3.7 (404)	12.6 (3619)	2.6	1.7 (1779)	<u>2.5</u>	3.0	—	× 5.0
GPQA	Acc.	39.9	43.0	—	46.9	51.39	<u>49.75</u>	—	↑ 8.4
	Token	6435	22450	—	3395	<u>3752</u>	3964	—	↓ 83.3%
	Time	102.7 (813)	358.4 (6108)	—	54.2 (798)	<u>59.9</u>	61.5	—	× 6.0
GSM8K	Acc.	81.1	90.4	<u>91.58</u>	89.5	91.28	92.34	—	↑ 1.9
	Token	1280	2370	<u>688</u>	353	704	932	—	↓ 70.3%
	Time	7.5 (449)	13.8 (1365)	<u>4.1</u>	2.1 (702)	4.4	4.8	—	× 3.1
HumanEval+	Acc.	74.4	76.8	82.62	78.0	83.84	<u>83.23</u>	—	↑ 7.0
	Token	2507	8768	1688	1274	1681	<u>1637</u>	—	↓ 81.3%
	Time	30.7 (502)	107.5 (1809)	20.2	15.6 (439)	<u>20.1</u>	21.0	—	× 5.3
MBPP+	Acc.	64.8	71.9	70.97	72.2	<u>72.75</u>	72.16	73.21	↑ 0.8
	Token	2053	7703	<u>1436</u>	1264	1502	1460	12768	↓ 81.0%
	Time	33.8 (1064)	126.8 (3898)	20.3	20.8 (1387)	20.1	<u>20.2</u>	92.2	× 6.3
MedQA	Acc.	53.0	<u>76.3</u>	75.33	77.0	73.33	72.67	—	↓ 3.0
	Token	2098	6893	1391	1007	1351	<u>1336</u>	—	↓ 80.6%
	Time	25.1 (476)	82.5 (3387)	15.5	12.1 (964)	15.7	<u>14.8</u>	—	× 5.6

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 3: Sequential MAS results for Qwen3-14B. Acc. is accuracy (%), Single, TextMAS, and Linear times are converted to the Kernel machine’s throughput; original A100 Speed values are shown in parentheses. The remaining methods retain their H200 measurements. Best results in each row are bold and second-best results are underlined. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by –.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
Sequential MAS Setting									
AIME24	Acc.	63.3	63.3	—	66.7	72.08	69.58	—	↑ 8.8
	Token	11263	32092	—	10593	8259	8489	—	↓ 74.3%
	Time	348.6 (1018)	993.2 (4554)	—	327.8 (1149)	255.6	283.4	—	× 3.9
AIME25	Acc.	56.7	60.0	—	63.3	56.67	60.00	49.58	↑ 0.0
	Token	11298	44618	—	11402	10266	10402	21403	↓ 77.0%
	Time	369.2 (1040)	1458.1 (5184)	—	372.6 (1473)	335.5	350.6	575.5	× 4.3
ARC-C	Acc.	92.6	95.9	95.99	95.6	95.48	95.73	—	↓ 0.2
	Token	773	2985	463	426	461	544	—	↓ 84.6%
	Time	7.9 (338)	30.4 (5125)	4.3	4.3 (1136)	4.3	6.0	—	× 7.1
ARC-E	Acc.	97.2	99.0	98.48	99.4	98.82	98.57	—	↓ 0.2
	Token	608	1670	379	224	380	462	—	↓ 77.3%
	Time	4.3 (551)	11.8 (9171)	2.8	1.6 (2124)	2.7	4.5	—	× 4.4
GPQA	Acc.	48.5	51.5	—	52.0	55.81	54.42	53.54	↑ 4.3
	Token	5547	12676	—	5454	3009	2995	18528	↓ 76.4%
	Time	109.5 (1043)	250.2 (9714)	—	107.7 (1475)	59.4	67.2	222.9	× 4.2
GSM8K	Acc.	83.7	93.8	92.12	95.2	93.10	93.33	—	↓ 0.5
	Token	1118	3324	577	644	571	643	—	↓ 82.8%
	Time	11.1 (536)	33.1 (3729)	5.0	6.4 (1952)	4.7	6.7	—	× 7.0
HumanEval+	Acc.	76.8	81.1	86.89	86.5	88.11	86.43	90.70	↑ 7.0
	Token	2366	5934	1590	2042	1594	1607	19587	↓ 73.1%
	Time	31.8 (1084)	79.7 (4062)	24.1	27.4 (1285)	24.1	28.1	157.3	× 3.3
MBPP+	Acc.	68.5	72.8	78.17	75.7	78.11	79.43	—	↑ 6.6
	Token	1858	4971	1345	1621	1336	1346	—	↓ 73.1%
	Time	28.5 (2410)	76.2 (8728)	21.9	24.9 (2400)	21.5	25.5	—	× 3.5
MedQA	Acc.	64.7	80.3	81.00	80.7	81.67	79.33	79.00	↑ 1.4
	Token	1746	3444	1616	1841	922	912	22875	↓ 73.5%
	Time	21.1 (1360)	41.6 (4142)	19.0	22.2 (1420)	13.2	14.3	151.0	× 3.2

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 4: Hierarchical MAS results for Qwen3-14B. Acc. is accuracy (%), Single, TextMAS, and Linear times are converted to the Kernel machine’s throughput; original A100 Speed values are shown in parentheses. The remaining methods retain their H200 measurements. Best results in each row are bold and second-best results are underlined. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
<i>Hierarchical MAS Setting</i>									
AIME24	Acc.	63.3	70.0	—	73.3	69.17	69.58	<u>71.25</u>	↓ 0.4
	Token	11263	29025	—	10230	8947	<u>9017</u>	23924	↓ 69.2%
	Time	357.6 (1018)	921.7 (5718)	—	324.8 (1089)	284.1	<u>291.0</u>	639.9	× 3.2
AIME25	Acc.	56.7	66.7	—	66.7	<u>59.17</u>	57.92	—	↓ 7.5
	Token	11298	50003	—	9527	<u>10755</u>	11122	—	↓ 78.5%
	Time	364.4 (1040)	1612.8 (6019)	—	307.3 (1056)	<u>346.9</u>	365.7	—	× 4.6
ARC-C	Acc.	92.6	95.3	<u>95.65</u>	95.5	95.99	<u>95.65</u>	—	↑ 0.7
	Token	773	2167	434	295	<u>424</u>	477	—	↓ 80.4%
	Time	6.5 (338)	18.1 (4283)	4.0	2.5 (1090)	<u>3.7</u>	5.4	—	× 4.9
ARC-E	Acc.	97.2	98.3	<u>98.70</u>	<u>98.7</u>	98.78	98.23	—	↑ 0.5
	Token	608	2752	<u>344</u>	619	337	402	—	↓ 87.8%
	Time	4.3 (551)	19.6 (7102)	<u>2.5</u>	4.4 (1884)	2.4	4.0	—	× 8.2
GPQA	Acc.	48.5	52.0	—	53.0	<u>54.80</u>	55.81	54.55	↑ 3.8
	Token	5547	20931	—	3606	<u>2769</u>	2755	20909	↓ 86.8%
	Time	123.2 (1043)	464.9 (9119)	—	80.1 (1458)	61.5	<u>63.6</u>	287.8	× 7.6
GSM8K	Acc.	83.7	90.8	<u>91.96</u>	91.6	91.81	92.49	—	↑ 1.7
	Token	1118	3021	574	495	<u>569</u>	602	—	↓ 81.2%
	Time	10.9 (536)	29.4 (3675)	4.6	<u>4.8 (1631)</u>	4.6	6.0	—	× 6.4
HumanEval+	Acc.	76.8	84.1	<u>88.41</u>	86.6	87.04	86.74	91.16	↑ 2.9
	Token	2366	8114	1473	1512	<u>1454</u>	1430	21403	↓ 82.4%
	Time	26.4 (1084)	90.6 (3988)	<u>23.8</u>	16.9 (1188)	24.0	25.4	185.6	× 3.8
MBPP+	Acc.	68.5	73.0	76.92	73.8	<u>78.04</u>	78.31	—	↑ 5.3
	Token	1858	7458	1209	1187	1132	<u>1182</u>	—	↓ 84.8%
	Time	<u>16.3 (2410)</u>	65.4 (9162)	23.3	10.4 (2507)	21.6	23.7	—	× 3.0
MedQA	Acc.	64.7	78.0	81.33	78.3	<u>79.00</u>	77.67	48.33	↑ 1.0
	Token	1746	5473	<u>870</u>	899	860	875	23700	↓ 84.3%
	Time	31.6 (1360)	99.0 (7591)	12.1	16.3 (1250)	<u>12.4</u>	13.8	173.7	× 8.0

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.